

Powertrain Control Valve Rebuild Strategy

Application Maintenance **Component
Rebuild** Safety MARC
Management

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1.0 Introduction

Control valves continue to be a critical component for long rebuilt torque converter and transmission service lives. The same technician who rebuilds the clutches, planetary and transfer gear groups typically handles control valve rebuilding – and in the same location.

This Control Valve Rebuild Strategy Best Practice involves separating the valve rebuild process from the other major component activities. It means treating valve rebuilding in a similar environment to diesel engine fuel system service.

The valve group should be transferred from the general disassembly and cleaning area to a “clean/controlled” environment for rebuild by a specialist. Then after rebuild it is returned to the assembly area for installation.

This rebuild strategy:

- Improves valve rebuild consistency and efficiency.
- Reduces the power train rebuild technician’s skill requirements in the general rebuild and assembly functions.
- Improves rebuild quality with potential to improve component life.

2.0 Best Practice Description

Valves are the most complex group in the major component. A typical rebuild technician processes only one valve group per rebuild job, giving him limited repetitive exposure to valve rebuilding. This situation:

- Reduces efficiency.
- Increases error risk.
- Increases technician skill requirements and cost.

Valve parts are more vulnerable to debris than other component parts and require special cleaning. They are typically hand cleaned in the washers used for other parts.

Valve handling damage and/or poor reuse decisions often result in unnecessary replacement and/or mid-life failures. Heavy debris-related damage can make the root cause of failure difficult to determine.

Valve groups are complex assemblies that are best handled by an experienced specialist. Experience is best developed with job repetition. Repetition may not sufficiently occur if the technician rebuilds the entire major component. It is recommended that a special technician handle the valve group rebuild.

Once removed from the major component, the valve groups must first be transferred to a specialized valve rebuild area.

Note: Some dealers use this area to rebuild other machine-related valves, such as implement, and/or steering clutch and brake valves.

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The valve rebuild area should be a climate-controlled area with filtered air inlets that create a positive air pressure to keep other shop dust/debris out. This area may be similar to traditional fuel system rebuild area recommendations.

The valve rebuild area is purpose-designed with:

- Valve-specific wash baskets to organize/protect disassembled parts
- Automatic washing equipment for consistent cleanliness, no damage, and reduced rebuild labor.
- Special tools for high-quality work and reduced rebuild labor.
- Effective valve group packaging and contamination control supplies to protect the group until installed on the major component.
- Protective staging area where the group will remain until the general assembly technician is ready to install it into the major component.

The valve group is transferred to the major component general assembly bay when the assembly technician is ready for it. Staging it in this bay prior to installation is acceptable only if the valve is fully assembled and protected from contamination and external physical damage.

3.0 Implementation Steps

1. Determine the anticipated type and volume of valves to be rebuilt annually and the forecast for future volumes for at least the next five years.
 - a. Determine the total number of labor hours required to perform valve rebuilds and the number of employees required to supply these hours.
 - b. Establish the amount of space required to disassemble, clean, rebuild, and assemble the valves.
 - c. Establish what handling/storage equipment is required to stage incoming, in-process, and outgoing valve groups.
 - d. Include peak factors to accommodate for expected business cycles, overall increases/decreases, and sudden transients as expected.
 - e. Include specialized product storage (shelves, carts, racks, etc.) and handling (cranes, carts, lift trucks, tugs, etc) equipment.
2. Establish tooling needs and storage requirements.
 - a. Research current dealer tool inventories and rebuild references/notes.
 - b. Research Caterpillar tool recommendations and rebuild references.
 - c. Establish needed cleaning, rebuild, and storage equipment for groups and tools.
3. Apply the strategy to shop layout plans.
 - a. Consider rebuild location's effects on product protection and flow paths.
 - b. Include shop technician teams in the review/application stages.
4. Consider the current segregated fuel system rebuild area. This area may have reduced utilization with the advent of electronic and/or unit fuel injector applications.
5. Develop, adjust and document rebuild procedure with quality checklists as needed.
6. Establish new/adjusted repair/rebuild time requirement targets as needed.
7. Install the layout/equipment changes into the shop (This may occur in phases as needed.)
8. Train employees to use the new procedures/equipment. Follow-up/enforce immediately.
9. Review process/equipment performance once established.
10. Establish and install any adjustments as needed.

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4.0 Benefits

- Increased capacity: These concepts reduce space/time requirements in the general rebuild bays, allowing the dealer to increase the number of jobs completed in each bay.
- Increased space utilization: Fuel rebuild area is often underutilized due to new product designs. Unused space could be used for valve rebuilding.
- Increased quality: Valve rebuild technicians develop more specialized skills in the most complex rebuild area, increasing rebuild service life
- Reduced labor/costs: Dealer technicians work more effectively with more space, specialized support equipment/tools, and fewer job variables. Valves are rebuilt faster.
- Technicians require fewer skills: Specialized jobs allow dealers to develop/utilize less experienced technicians sooner after hire date. They can also develop them effectively with increasing skill requirements. Technicians do not have to know everything to contribute to a rebuild process.

5.0 Resources Required

- Facility
 - Re-organize workstation/shop to allow for valve rebuild process, incoming and outgoing staging, parts washer, and a workbench as needed.
 - The fuel rebuild area may support this function with little resource adjustment, depending on its current status, size, and rebuild volume.
- Support Equipment
 - Well-built washers will cost approximately US\$2,000 - US\$8,000.
 - A generic workbench to disassemble/assemble valves as needed.
 - Specialized Caterpillar tools. (The dealer may already have these.)
 - Staging equipment/racks as needed for incoming/outgoing valves. Include containers for valve protection. Eliminates floor staging.
 - Lifting devices such as cranes may be shared with other functions in the shop. Lift requirements are relatively low for a typical Caterpillar repair/rebuild facility and would be determined by the size of valve groups being handled.
 - Lifting chains, straps, and/or brackets may also be shared with other functions.
- People
 - Establish current valve rebuild concerns with team
 - Establish process, layout, and equipment improvements through team.
- Training
 - Define/establish/enforce new processes, roles, and responsibilities with production teams.

6.0 Supporting Attachments / References

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Enclosed dedicated room for pump and valve rebuilds – MATCO CRC in Hermosillo, Sonora, MX

7.0 Related Best Practices

- 1009-4.01-1211 - Transmission Control Valve Test Bench

8.0 Acknowledgements

- This Best Practice was written by:
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